

Exp.NO 3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss.

OBJECTIVE 1 :

To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models.

CODE:

```
clc; clear; close all;

d = 0.1:0.1:5; % Distance (km)

f = [900 3500 28000]; % Frequency (MHz)

PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);

figure

% ----- Graph 1 -----
subplot(2,1,1)

plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)

title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')

grid on

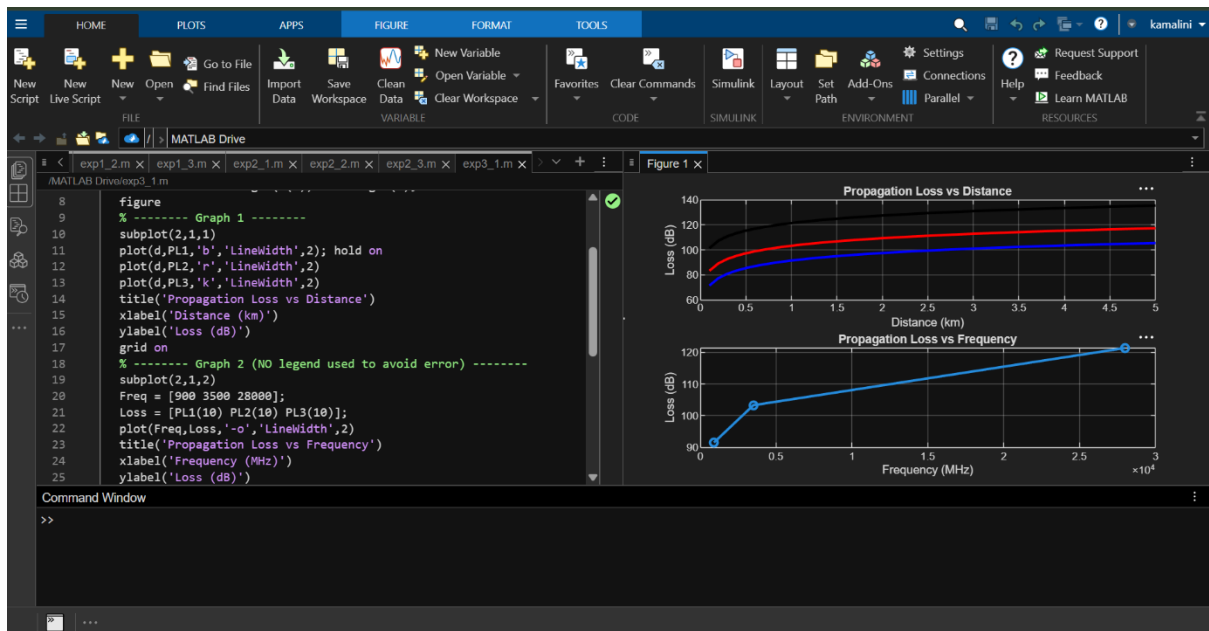
% ----- Graph 2 (NO legend used to avoid error) -----
subplot(2,1,2)

Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];

plot(Freq,Loss,'-o','LineWidth',2)

title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')

grid on
```



OBJECTIVE 2 :

To evaluate the impact of different Operating Frequencies (GHz) on signal attenuation and coverage performance.

CODE:

```
clc; clear; close all;
```

```
d = 2; % fixed distance (km)
```

```
f = [900 3500 28000]; % MHz
```

```
PL = 32.44 + 20*log10(f) + 20*log10(d);
```

```
figure
```

```
% ----- Graph 1: Frequency vs Loss -----
```

```
subplot(2,1,1)
```

```
plot(f,PL,'-o','LineWidth',2)
```

```
title('Propagation Loss vs Operating Frequency')
```

```
xlabel('Frequency (MHz)')
```

```
ylabel('Propagation Loss (dB)')
```

```
grid on
```

```
% ----- Graph 2: Band Comparison -----
```

```
subplot(2,1,2)
```

```
bar(PL)
```

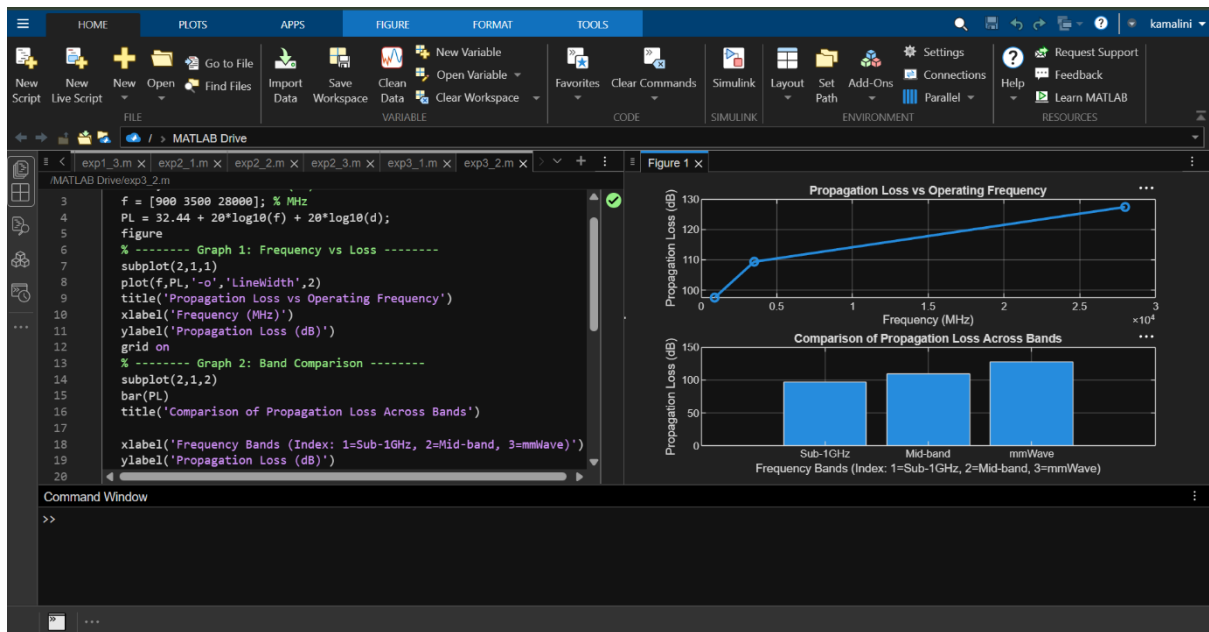
```
title('Comparison of Propagation Loss Across Bands')
```

```
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
```

```
ylabel('Propagation Loss (dB)')
```

```
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
```

```
grid on
```



OBJECTIVES 3 :

To analyze how Communication Distance (km) affects propagation loss across different spectrum bands.

CODE:

```
clc; clear; close all;  
  
% Frequency bands (MHz)  
f = [900 3500 28000];  
  
% Distance (km)  
d = 2;  
  
% Propagation Loss (FSPL)  
PL = 32.44 + 20*log10(f) + 20*log10(d);  
  
figure  
  
% ----- Graph 1: Simple Line Plot (SAFE) -----
```

```
subplot(2,1,1)
```

```
plot(f, PL, '-o', 'LineWidth', 2)
```

```
title('Propagation Loss vs Operating Frequency')
```

```
xlabel('Frequency (MHz)')
```

```
ylabel('Propagation Loss (dB)')
```

```
grid on
```

```
% ----- Graph 2: Safe Comparison Plot -----
```

```
subplot(2,1,2)
```

```
plot(1:3, PL, '-s', 'LineWidth', 2)
```

```
title('Band-wise Propagation Loss Comparison')
```

```
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
```

```
ylabel('Propagation Loss (dB)')
```

```
grid on
```

